

Quantum Toom's Contours

David Ponarovsky

March 8, 2026

Introduction

Today:

- ▶ Proof.
 1. Walkthrough the math

Our Code

Code

Consider the representation of the d -dimensional toric by the cayley graph of the $(\mathbb{Z}_n^d, +)$ group generated by the elements $S = \{1_i : i \in [d]\}$ ¹.

In our code, bits are set on the plaquette, and the edges correspond to checks that restrict the parity of the bits on their support to be even.

¹where 1_i is the vector consisting of one at the i th coordinate and zero elsewhere

Local Decoder

Positive edge

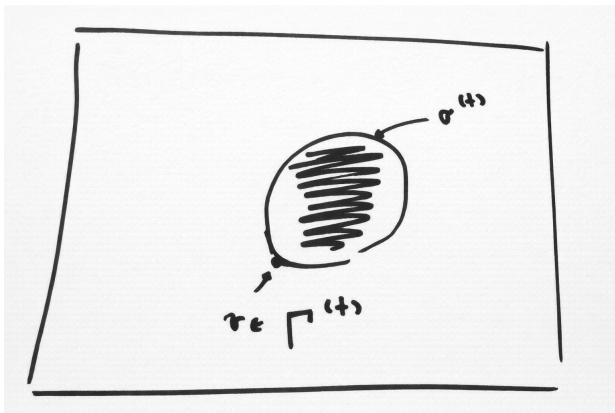
We say that an edge $\{x, y\}$ is positive relative to a plaquette $\{v, g_1 v, g_2 g_1 v, g_2 v\}$, defined by a rooted vertex $v \in \mathbb{Z}_n^d$ and the generators $g_1 \neq g_2 \in S$, if $x = v$ and y is either $g_1 v$ or $g_2 v$.

Toom's rule

We generalize Toom's rule in the following manner: Each plaquette asks if both of its positive edges point to a syndrome, namely unsatisfied. If so, then the bit flips itself; for convenience, we will think of the vertex at the end of the positive edges as the vertex which makes the decision and flips the plaquette.

Title

Let $\sigma^{(t)}$ be a domain wall at time t , and let $\Gamma^t = \{v : \exists \{v, u\} \in \sigma^t\}$.



u Is Added By v

We denote by ϕ^t the subsets of bits which are flipped as a result of applying the Toom's rule at time t , and by ϕ_v^t the bits which are flipped by vertex v .

We say that u is added to Γ^{t+1} by v at time t , if $u \in \Gamma^{t+1}$ and $u \in V(\phi_v^t)$. When it clear form the context, we will omit the time, and say that u is added by v .

Potentials Functions

Define the potentials functions $L_i : \mathbb{R}^d \rightarrow \mathbb{R}$ as follow:

$$L_i(\mathbf{x}) = \begin{cases} \mathbf{x}_i & i < d + 1 \\ -\langle \mathbf{1}, \mathbf{x} \rangle & i = d + 1 \end{cases}$$

The **Size** of a subset S is defined as:

$$\text{Size}(S) = \sum_i \max_{\mathbf{x} \in S} L_i(\mathbf{x})$$

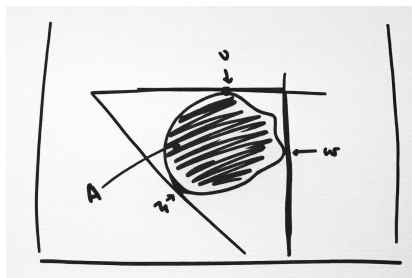


Figure: Example in 2D. A is a flipped droplet, $L_x(A) = \max_{v \in \Gamma^t} v_x = w$.

Title

claim

Let $v \in \Gamma^t$ for any u that is added by v , and for any $x \in [d+1]$, we have $L_x(u) \leq L_x(v) + 1$.

proof

By definition, any u that is added by v is of the form $v + \sum \alpha_j 1_j$, where $\alpha_j \in \{0, 1\}$ and $\sum \alpha_j \leq 2$. Thus, for any $x \in [d]$:

$$L_x(u) = \langle 1_x, v + \sum \alpha_j 1_j \rangle = \langle 1_x, v \rangle + \alpha_x \leq L_x(v) + 1$$

For $x = d+1$ we have:

$$L_x(u) = \langle -\mathbf{1}, v + \sum \alpha_j 1_j \rangle = \langle -\mathbf{1}, v \rangle - \sum_j \alpha_j \leq L_x(v)$$

claim

Let $v \in \Gamma^t$ such that $L_{d+1}(v)$ is a (local) maxima in Γ^t then for any u which is added by v we have $L_{d+1}(u) \leq L_{d+1}(v) - 1$.

Title

proof

We will prove that $v \notin \Gamma^{t+1}$, and then by repeating on ?? where $\sum \alpha_j \geq 1$, we get the desired result. Assume, toward contradiction, that $v \in \Gamma^{t+1}$, and denote by w the vertex that adds v . If $w \neq v$, then there is at least one coordinate i for which $w_i < v_i$, thus $L_{d+1}(w) > L_{d+1}(v)$, in contradiction to $L_{d+1}(v)$ being a maximum in Γ^t . In the other case, where v adds itself, it follows that σ_v can't be decomposed into pairs of unsatisfied checks; namely, $|\sigma_v|$ is odd. We show by induction that impossible, namely that $|\sigma_v|$ is always even. The induction is over the size of the plaquettes subset that induce the boundary σ .

1. Base. Consider a boundary that induced by a single plaquette, in that case it obvious that $|\sigma_v|$ is either zero (where v is not belong to the plaquette) or 2.
2. Assume for any boundary that induced by at most k plaquettes, and consider a boundary σ which is induced by $k + 1$ plaquettes, name this subset F and observes that F can

claim

Let $x \in [d]$ and let $v \in \Gamma^t$ such that $L_x(v)$ is a (local) maxima in Γ^t then for any u which is added by v we have $L_x(u) \leq L_x(v)$.

proof

Since $L_x(v)$ is maximal, v doesn't see edges that goes from it at the positive x -direction, otherwise denote it by e and denote by w the second vertex at the end of e , w has an x -coordinate which is greater than v 's x -coordinate by 1 and in addition $w \in \Gamma^t$ (since it lays on the support of e). Therefore $L_x(w) > L_x(v)$ in contradiction to $L_x(v)$ be maximal in Γ^t . Hence, any plaquette that is being flipped by v lies on edges associated with 1_j such that $j \neq x$. Thus, any vertex u that is added by v has the form $v + \sum_{j \in [d] \setminus \{x\}} \alpha_j 1_j$, where $\alpha_j \in \{0, 1\}$ and $\sum \alpha_j \leq 2$. So:

$$L_x(u) = \langle 1_x, v + \sum_{j \in [d] \setminus \{x\}} \alpha_j 1_j \rangle = \langle 1_x, v \rangle = L_x(v)$$

claim

Let Γ^t be the vertices in the support of the domain wall σ^t .

Denote $v_i = \arg \max_{v \in \Gamma^t} L_i(v)$. Then for any $i \in [d]$, one can find $u \in \Gamma^{t-1}$ such that u adds v_i and $L_i(u) \geq L_i(v_i)$. In addition, one can find $u_{d+1} \in \Gamma^{t-1}$ such that u_{d+1} adds v_{d+1} and $L_{d+1}(u_{d+1}) = L_{d+1}(v_{d+1}) + 1$.

Title

proof

Just take u_i to be any vertex that adds v_i . By claims ?? , ?? , and ?? , we get the claim.

Title

Title